

Research article

ESG greenwashing and organizational resilience: Exploring sustainable development paths from the perspective of unsystematic risk

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ABSTRACT

Amid accelerating globalization, deepening green transitions, and digital upheavals, corporate organizational resilience has emerged as an essential element for economic resilience, with genuine ESG practices at its core. However, intentional distortions in ESG implementation, particularly pervasive greenwashing, pose major risks by eroding adaptive capacities, distorting market signals, and misallocating resources, ultimately impeding high-quality economic development. This study examines 2012–2022 data from China's A-share listed companies through the dual lenses of unsystematic risk transmission and governance failures, revealing how ESG greenwashing undermines organizational resilience by reducing competitiveness, tightening financing constraints, and stifling innovation. Evidence suggests that green management innovations and analyst attention reduce these effects by optimizing internal risk controls and increasing transparency. Research on organizational heterogeneity reveals that non-SOEs, tech-driven, and growth-stage firms with robust internal controls demonstrate stronger regulatory resistance through structural agility. Paradoxically, rigorous environmental regulations may counterproductively incentivize ESG greenwashing that erode dynamic capabilities. This study provides actionable insights for safeguarding environmental integrity and advancing sustainable development through a risk management framework to counteract compliance circumvention strategies.

1. Introduction

Establishing a robust ESG (environmental, social, and governance) ecosystem with Chinese characteristics is critical for advancing high-quality economic development and the Beautiful China Initiative. Nowadays, the intensifying impacts of climate change—from physical risks like extreme weather events and ecological degradation to transition risks such as low-carbon technology disruption and carbon border tariffs—are creating systemic threats to sustainable growth. In addition, compounded by VUCA characteristics (volatility, uncertainty, complexity, and ambiguity) (Bennett and Lemoine, 2014) of the new globalization, the global and systemic risks triggered by fluctuations in the macro environment will continuously break through the vulnerability threshold of economic and social development. So enterprises, as key actors in resource allocation and environmental governance, must strengthen risk management frameworks and dynamic adaptive capacities to stabilize microeconomic foundations and enhance macroeconomic resilience.

The ESG ecosystem creates a collaborative network among

governments, businesses, investors, and civil society that can systematically evaluates environmental performance, internalizes social responsibility, and operationalizes governance effectiveness. Enterprises, embedding ESG principles within risk systems, can optimize their resource allocation techniques (Wu and Chen, 2025) and detect sustainability issues more accurately (Wang et al., 2024a), which will improve organizational adaptability (Ma et al., 2024). However, compared with the West, China's ESG practice is still lagging, and the corporate environmental information disclosure system has problems such as vague standards, incomplete coverage, and a lack of verification. Some enterprises avoid substantive rectification by selective disclosure of environmental protection cases, dummy CSR projects (It refers to the corporate social responsibility projects that enterprises fabricate but do not actually implement in order to enhance their own image or meet certain specific needs, e.g., Fabricating low-carbon R&D projects, overestimating environmental protection investment or emission reduction capabilities) and other ESG greenwashing methods. It will intensify the unsystematic risks (idiosyncratic risks) within the enterprise and weaken the organization's risk perception and emergency

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response abilities. As a result, developing an effective mechanism for identifying and mitigating ESG greenwashing risks is critical for preserving company green integrity and increasing overall organizational resilience. It is also the foundation for ensuring that the ESG ecosystem runs smoothly. This study deconstructs the inherent relationship between ESG greenwashing and organizational resilience to better identify, assess, and respond to ESG greenwashing threats.

This study identifies and evaluates enterprises' unsystematic risks triggered by ESG greenwashing from a risk management perspective. By integrating internal-external perspectives through green management innovation and analyst attention, this study explores effective countermeasures for mitigating such risks. Furthermore, the research meticulously examines the heterogeneous effects of ESG greenwashing on organizational resilience across different contexts, revealing its context-dependent characteristics. In addition to enriching existing literature, the analysis offers empirical insights at the micro level for controlling ESG greenwashing and boosting organizational resilience in concert. These findings have important theoretical and practical ramifications to help businesses standardize ESG disclosures, fight ESG greenwashing, and improve organizational agility.

Compared with existing literature, this study mainly makes the following contributions. Firstly, it contributes to the literature, which includes ESG greenwashing and corporate organizational resilience in an analytical framework and reveals the potential negative impacts of ESG greenwashing on organizational resilience, providing empirical evidence about ESG greenwashing risks for enterprises and society.

Secondly, this study integrates stakeholder, expectancy, signaling, and dynamic capability theories to reveal how ESG greenwashing undermines organizational resilience through dual mechanisms: unsystematic risk transmission and governance failure. Specifically, we demonstrate its erosion pathways, including weakening competitive advantages, exacerbating financing constraints, and impairing innovation capacity. Furthermore, the research proposes actionable governance frameworks for enterprises to establish risk early-warning systems and implement management innovations.

Thirdly, this study deconstructs organizational resilience into granular components using multidimensional heterogeneity analyses and utilizes the panel quantile regression model to analyze the evolution trajectory of the marginal effect of ESG greenwashing in the process of organizational resilience. By moving beyond conventional, siloed governance approaches, this study develops an integrated governance model encompassing firm-specific characteristics identification, external environment alignment, and digital technology integration. Finally, we propose targeted governance measures and policy recommendations to enhance enterprises' adaptive capacity against external shocks and create novel pathways for sustainable development.

The rest of this study is organized as follows. Section 2 surveys the literature. Section 3 describes theoretical analysis and hypotheses. Section 4 explains the empirical model design, data source, and variables. Section 5, Section 6, and Section 7 present empirical analysis, while Section 8 includes conclusions and implications of the research.

2. Literature review

2.1. The connotation and influencing factors of enterprise organizational resilience

Resilience, as a cross-disciplinary research concept, has its theoretical roots in physics and ecology (Holling, 1973). With the continuous transformation of the global economic landscape and the increasing uncertainty, the research on resilience theory in economics and management has shown systematic and hierarchical evolution characteristics. Macro-level resilience reflects an economic system's capacity to resist, adapt to, and recover from external shocks (Nazari et al., 2025), while micro-level analysis focuses on enterprises' sustainable development capabilities through predictive, adaptive, and risk-mitigation

strategies (Williams et al., 2017) (Teece, 2007). Under the influences of intensified climate change, frequent natural disasters, and escalating geopolitical conflicts, corporate organizational resilience not only determines the survival and development of micro-subjects but also, through the transmission effect of industrial networks and market mechanisms, converges to form the overall shock resistance and post-crisis recovery potential of the macroeconomy. This micro-macro transmission mechanism underscores enterprise resilience as a breakthrough point for strengthening systemic economic resilience.

Dynamic capability theory conceptualizes organizational resilience as an enterprise's capacity to continuously integrate and reconfigure resources in complex environments (Teece, 2007) (Heikinheimo et al., 2025). This capability has multi-dimensional functions. In risk perception, organizational resilience enables enterprises to sensitively identify systemic risk signals (e.g., technological, institutional, or market shifts) and complete strategic resource reserves before the crisis emerges through establishing early warning systems and scenario planning (Krasnopevtseva et al., 2025). Regarding crisis response, it provides flexibility to maintain core operations via structural adjustments, process optimization, and resource reallocation (Li et al., 2025a). Finally, in recovery and development, organizational resilience not only supports enterprises' operational reconstruction but also drives strategic transformation post-crisis (Stoverink et al., 2018).

Currently, the influencing factors of organizational resilience have also been widely studied. On the one hand, organizational resilience will be influenced by macro-environmental disruptions and institutional changes. Exogenous risks like pandemic shocks, economic policy uncertainty (Ayadi and Abderrahman, 2025), and financial misallocation (Huang et al., 2024) may erode enterprises' viability via supply chain breakdowns or market contractions. However, some institutional changes, such as reform dividends (Guo et al., 2024) and supply chain diversification (Cohen et al., 2022) can provide them with strategic opportunities to improve capabilities. On the other hand, enterprises can construct micro-level internal capabilities through coordinated evolution, including technological empowerment and organizational change. Firms can enhance their operational efficiency through digital transformation, specifically via industrial robotics and cloud computing (Jiang et al., 2025), and improve their risk sensitivity through big data-driven risk warning systems (Xiao and Xie, 2024). Simultaneously, strategic renewal (Liang and Li, 2024), business model innovation (Gómez et al., 2024), and ESG practices (Wu and Chen, 2025) can help them restructure resource allocation and accelerate high-quality transformation. Additionally, firms with high ESG performance usually have stronger sustainability. Firstly, enterprises are regulated by ESG information disclosure, which will prompt them to reduce carbon emissions (Saha and Maji, 2025) and inefficient investments (Liu, 2025). Secondly, through improving ESG performance, enterprises can not only reduce market volatility (Naseer et al., 2024) and bankruptcy risks (Wu and Chen, 2025) (Chen et al., 2025) but also avoid taxes (Yuan et al., 2025) and increase long-term value. Therefore, a higher enterprise's ESG score will be regarded as indicating stronger adaptability and resource allocation capabilities. Moreover, some researchers have found that ESG practice has non-linear relationships and spillover effects on enterprise performance and sustainability (Bagh et al., 2024a) (Ren et al., 2025). In conclusion, under the background of the ESG ecosystem reconstructing the enterprise risk governance paradigm, the formation mechanism of enterprise organizational resilience urgently needs to be re-examined within the framework of ESG practices.

2.2. The causes and economic consequences of ESG greenwashing and unsystematic risks

Enterprise risks comprise systematic risks and unsystematic risks. Systematic risks are non-diversifiable and arise from market environment or macroeconomic factors (Huang et al., 2021). Unsystematic risks (idiosyncratic risks) are related to specific assets or investment

portfolios and can be diversified (Huang et al., 2021), which is mainly triggered by specific factors of the enterprise itself or the industry, such as decision-making mistakes, governance deficiencies, technological shocks, and competitive changes. Although extensive research concludes that ESG practices can effectively reduce firms' idiosyncratic risks and enhance their sustainability (Ren et al., 2025) (Merladet et al., 2025) (Li et al., 2025b), the potential alienation behavior in ESG practices (ESG greenwashing), manifested through strategic fiction, selective disclosure, or exaggerated promotion (Yu et al., 2020), has not received sufficient attention. Such organizational deviance behaviors, aimed at securing benefits, may amplify firms' unsystematic risks, including credit risks, business risks, and financial risks, by introducing new uncertainties.

ESG greenwashing is a complex socio-economic phenomenon, arising from multifaceted internal and external drivers. Internally, because of its long-term horizon, high uncertainty, and low short-term returns, ESG investment may conflict with enterprises' immediate financial goals. In addition, ESG practices are constrained by rigid conditions such as technical capabilities, resource endowments, and professional talent reserves (Wang and Zhang, 2024). Sometimes, they might drain firms' operational funds without yielding direct economic benefits (Ma et al., 2023), which will force management to navigate performance pressures versus sustainability demands and induce them to engage in ESG greenwashing (Zhao and Wang, 2025). Externally, ESG greenwashing can offer firms rapid market rewards through image enhancement (e.g., reducing financing costs, attracting investments, and boosting brand premiums). Therefore, firms always regard greenwashing as a strategic "low-cost, high-return" response, particularly amid stakeholder scrutiny, consumer expectations, and fluctuating regulatory pressures. Besides, the current problems of information asymmetry in ESG evaluation systems, including inconsistent rating standards, opaque information disclosure, and the absence of verification, will undermine investor judgment and increase information processing costs and agency problems (Liu et al., 2024), objectively creating institutional loopholes for ESG greenwashing.

Defined as organizational misconduct involving strategic fabrication, selective disclosure, or exaggerated promotion of environmental and social governance performance (Yu et al., 2020) (Liu et al., 2024) (Daga et al., 2025), ESG greenwashing has become a critical governance challenge due to its complexity, professional opacity, and delayed consequences (Sneideriene and Legenzova, 2025). Existing research shows that while ESG greenwashing may temporarily improve financial performance through reputational premiums and reduced financing costs (Lee and Raschke, 2023) (Parguel et al., 2015), its long-term negative repercussions are widely acknowledged (Sneideriene and Legenzova, 2025) (Sajid et al., 2024). Operationally, greenwashing can create compounded risks through reputational capital erosion (Delmas and Burbano, 2011), strategic partnership decoupling (Shen et al., 2023), and debt accumulation (Liu et al., 2022). In terms of decision-making, enterprises' short-term performance emphasis can promote route dependency, which may stifle their significant innovation investments (Lian et al., 2022) and strategic transformation (Zhang et al., 2025a). In the governance dimension, intensified agency conflicts may distort firms' ESG resource allocation, widening the gap between pseudo-green investments and genuine transition needs (Yu et al., 2020). Besides, ESG greenwashing with "low-cost, high-return" nature under competitive mimicry can exacerbate information asymmetry (lemons market effects) (Li et al., 2023), distort carbon pricing signals (Ye and Huang, 2025), undermine resource allocation efficiency (Lee and Raschke, 2023), and weaken policy enforcement (Zhang, 2023a). These mechanisms collectively trigger market failures, obstructing the green transition and high-quality development. Thus, it is vital for enhancing organizational resilience and ensuring sustainable economic progress to effectively address ESG greenwashing.

Notably, through the above review, it is found that existing research on organizational resilience has predominantly focused on identifying

macro-systemic risks and building defensive mechanisms, yet lacks in-depth analysis of micro-level unsystematic risk transmission and its disruptive impacts. Furthermore, though ESG greenwashing has emerged as a novel source of organizational vulnerability, current studies remain largely theoretical in exploring its negative effects, failing to comprehensively reveal how such practices undermine organizational resilience through the covert channel. Additionally, important questions remain amidst escalating environmental challenges and proliferating national sustainability initiatives: Through what channels does ESG greenwashing operate? What evolution trend does the marginal effect of ESG greenwashing show as the level of organizational resilience increases? Crucially, how can adaptive governance frameworks reduce greenwashing concerns and improve ESG's resilience-enhancing potential? Addressing these gaps holds theoretical significance for advancing resilience theories and practical urgency for guiding sustainable transitions.

3. Theoretical framework and hypotheses

Corporate organizational resilience is one of the decisive factors for companies to maintain a competitive advantage in adversity (Lengnick-Hall et al., 2011). However, various forms of greenwashing derived from ESG practice can significantly increase the probability of idiosyncratic risk and weaken the firm's ability to adjust dynamically (Baldi and Pandimiglio, 2022). Based on stakeholder theory, expectation theory, signaling theory, and dynamic capability theory, this research offers a thorough examination of how ESG greenwashing affects organizational resilience.

3.1. ESG greenwashing and corporate organizational resilience

While organizational resilience relies on its intrinsic developmental capacities, it is also dependent on the collaborative exchanges between diverse economic entities. Stakeholder theory suggests that firms maintain symbiotic value relationships with stakeholders, necessitating a balanced consideration of shareholder value maximization and social responsibilities within resource constraints (Weng et al., 2025) (e.g., environmental protection, governance optimization, and community welfare). Although ESG investments can mitigate conflicts that erode resilience, their long-term horizons, high risks, and uncertainties (Zhang et al., 2022a) often incentivize firms to instrumentalize social responsibilities (Wang et al., 2023) and substitute ESG commitments with ESG greenwashing tactics. For instance, in product markets, firms may deploy "claimed greenwashing" (by making false or exaggerated textual statements, labels or certifications, misleading consumers into believing that their products or services have environmental protection attributes, but lacking substantive actions or evidence to support it) and "executional greenwashing" (creating environmental illusions through specific actions or visual designs wrongly leads consumers to associate environmental features with brands or products) to project green images (Parguel et al., 2015) (Pancer et al., 2017), leading to consumer premiums for short-term gains. In financing, firms secure low-cost capital through selective disclosures, inflated governance metrics, and misused sustainability labels, but divert funds to non-green projects (Bowen and Aragon-Correa, 2014). In regulatory contexts, firms manipulate emission data, forge certifications, and exploit political connections to feign compliance to evade environmental oversight costs (Liu et al., 2024).

The profit incentives of ESG greenwashing, reinforced by its diverse forms, operational complexity, and technological opacity, often lead managers to underestimate its risks or even foster a cognitive bias that risks will diminish over time (Ghanbarpour and Gustafsson, 2022). In reality, ESG greenwashing risks can persist and erode organizational resilience through multidimensional pathways across product markets, capital allocation, and regulatory compliance. Firstly, when enterprises engage in ESG greenwashing and are exposed, they inevitably face a trust crisis and plummeting brand value. This cascades into reduced

operational capacity, shrinking market share, and ultimately undermines both financial stability and competitive standing. Secondly, the misallocation of resources in ESG greenwashing will squeeze out the capital inflow and technological accumulation for substantive green transformation. This will continuously undermine enterprises' ability to deal with crises. Finally, as environmental enforcement increases and ESG disclosure standards tighten, enterprises with pseudo-green opportunism will face heightened compliance scrutiny and legal liabilities, which may escalate their regulatory costs and litigation risks. This shrinking regulatory arbitrage window may transform ESG greenwashing from short-term speculation into long-term liabilities. Consequently, this study proposes [Hypothesis 1](#).

H1. ESG greenwashing weakens organizational resilience.

3.2. ESG greenwashing, business risk, and corporate organizational resilience

Business risks permeate all operational phases. Organizational resilience, the capacity to mitigate these risks, enables firms to sustain stability and growth through strategic adaptation and resource optimization. ESG greenwashing, however, will damage the corporate image, weaken market power ([Liu et al., 2025](#)), and intensify business risks, thereby jeopardizing long-term corporate viability.

According to expectation theory, stakeholders and the public generally anticipate corporate responsibility in economic, social, and environmental domains. As sustainability awareness grows, stakeholders increasingly prioritize ESG practices and tend to set expectation benchmarks ([Shuang et al., 2024](#)). When corporate ESG performance underachieves against these societal benchmarks, it will provoke strong stakeholder reaction ([Luo and Su, 2025](#)) ([Xu et al., 2025](#)). Expectation benchmarks emerge through two mechanisms. First, stakeholders develop expectations based on corporate positioning. Consumers, for instance, rationally assess green product investments against anticipated environmental outcomes. Specifically, when consumers think the green claims made by an enterprise are trustworthy, the purchasing motivation of consumers will significantly increase. Conversely, when detecting the ESG greenwashing motives, the enterprise will experience negative emotions such as resistance and aversion ([Sajid et al., 2024](#)). What's worse, such negativity can create ripple effects ([Capelle-Blancard and Petit, 2019](#)), amplifying public exposure of both "claimed greenwashing" and "executorial greenwashing" ([Sajid et al., 2024](#)). Second, stakeholders frequently compare firms with industry peers. Companies lagging in social responsibility performance risk may lose product market share, fiscal subsidies, and financing access ([Zhu et al., 2022](#)), as these gaps signal deviation from societal expectations. As a result, the study proposes [hypothesis 2a](#).

H2a. ESG greenwashing weakens organizational resilience by exacerbating business risk.

3.3. ESG greenwashing, financial risk, and corporate organizational resilience

Signaling theory ([Connelly, 2011](#)) suggests that in markets with information asymmetry, well-managed firms with strong environmental performance are incentivized to disclose their true operational practices. They prefer to send positive signals to enhance market valuation, while underperforming firms often resort to selective disclosure strategies to gain legitimate benefits ([Clarkson et al., 2007](#)). The adoption of ESG rating policies has reinforced investor preference for firms with robust ESG performance, which are perceived as having lower idiosyncratic risk ([Tao et al., 2025](#)) and higher growth potential ([Bai et al., 2022](#)). However, information asymmetry ([Wang et al., 2024b](#)) creates opportunities for firms to engage in ESG greenwashing through impression management to secure financing. Such greenwashing practices, manifested in excessive welfare commitments and overinvestment, will

worsen firms' financing constraints, elevate their default probability, and intensify financial risks, which will ultimately enhance the enterprise's overall unsystematic risks ([Tao et al., 2025](#)) ([Zhang et al., 2024](#)).

On the one hand, when enterprises practice ESG greenwashing, they may misallocate their investment and financing resources. These behaviors can distort capital allocation and divert their finite investment resources away from high-value projects. In the long term, such misallocation will elevate default risks and trigger cascading effects, like reputational damage, liquidity stress, and debt accumulation, which amplify stock price crash risks. On the other hand, frequent defaults severely impair firms' credit ratings, prompting financial institutions to question their solvency, tighten credit policies, and raise financing thresholds. In this case, enterprises may lack sufficient financial flexibility and have difficulties in raising funds and adjusting strategies to face environmental changes and external shocks. Besides, such threats can also restrict their green innovation, thereby greatly weakening organizational resilience. Thus, the research proposes [hypothesis 2b](#) and [hypothesis 2c](#).

H2b. ESG greenwashing can exacerbate financial risk by increasing the probability of debt default, ultimately weakening organizational resilience.

H2c. ESG greenwashing can exacerbate financial risk by worsening financing constraints, ultimately weakening organizational resilience.

3.4. ESG greenwashing, innovation risk, and corporate organizational resilience

The uncertainties related to enterprise innovation will further lead to an increase in idiosyncratic risk ([Geeta et al., 2023](#)). Dynamic capability theory elucidates organizations' strategic adaptation mechanisms in complex environments ([Mensah et al., 2025](#)). The theory underscores that firms need to develop integrated capability systems encompassing environmental awareness, decision flexibility, and resource orchestration, to enhance responsiveness to technological shifts, regulatory changes, and market demands ([Heikinheimo et al., 2025](#)) ([Mensah et al., 2025](#)). It can help firms achieve dynamic alignment between organizational strategies and external ecosystems. Furthermore, enterprises can utilize this adaptive process to continuously improve organizational routines while fostering sustainable competitive advantages through innovation opportunity capture and strategic resilience development ([Teece, 2007](#)).

In the era of digitalization, the digital transformation of enterprises has emerged as a critical pathway for upgrading dynamic capabilities ([Chi et al., 2025](#)) ([Zhang et al., 2025b](#)). Through digital technologies and data-driven resource orchestration mechanisms, firms can improve their environmental sensing systems and strategic responsiveness to detect and respond to market signals more effectively and rapidly ([Yin et al., 2025](#)). However, the widespread adoption of big data has also transformed corporate greenwashing tactics, enabling some firms to exploit dynamic capabilities for regulatory arbitrage. Specifically, firms increasingly substitute strategic innovation for substantive R&D investments or prioritize the quantity of innovation over its quality, resulting in bubbles of high-quantity and low-quality innovation ([Tang and Huo, 2023](#)). Such tactical innovation not only erodes firms' foundation for technological accumulation and core capability development but also misleads their strategic decision-making through artificial competitiveness, trapping organizations in innovation paradoxes. Over time, the degradation of substantive innovation capabilities undermines sustainable competitive advantages, particularly for firms in transitional crises. This capability-hollowing phenomenon reinforces path dependency ([Li et al., 2020](#)), ultimately impairing long-term performance and the resilience of organizational survival. Therefore, this study proposes [Hypothesis H2d](#).

H2d. ESG greenwashing weakens organizational resilience by

exacerbating innovation risk.

3.5. The moderating role of green management innovations and analysts' concerns

The interplay of internal and external factors profoundly shapes how organizational resilience responds to ESG greenwashing (Yin, 2024), necessitating holistic and rigorous consideration of these aspects in pursuing sustainability goals.

Internally, proactive green management innovation is important. Enterprises that embed environmental and sustainability principles into core management systems can rapidly foster green innovation mindsets. This cultural shift enables the practical implementation of eco-conscious practices and the establishment of measurable, science-based sustainability targets (Bagh et al., 2024b) (Xi and Zhao, 2022). Concurrently, digital technologies can empower firms to build efficient governance systems (Wang and Hou, 2024), where eco-management tools can enhance energy efficiency, reduce emissions, and improve environmental performance. This not only mitigates social responsibility risks but also curbs managerial opportunism, bolstering adaptability to environmental uncertainties (Xia et al., 2020).

Externally, analyst attention serves as a vital governance method by improving information transparency and reducing asymmetries. Analysts can leverage expertise, data access, and technological tools to uncover and disseminate corporate information, exposing hidden ESG greenwashing practices that impact market valuations (Luo et al., 2024). Their oversight pressures management to enhance operational efficiency, optimize investments, and prioritize genuine sustainability over symbolic actions (Li and Liu, 2024). Collectively, internal green innovation and external monitoring create complementary safeguards, attenuating the adverse effects of ESG greenwashing on organizational resilience (Kong et al., 2013). Therefore, this study proposes the following hypotheses and outlines a risk management framework for ESG greenwashing impacts (Fig. 1).

H3a. Green management innovations reduce the negative effect of ESG greenwashing on organizational resilience.

H3b. Analyst attention reduces the negative effect of ESG greenwashing on organizational resilience.

4. Methodology

4.1. Sample selection and data sources

This study examines A-share listed firms in China from 2012 to 2022, applying the following data filters: (1) excluding samples under delisted status (ST, ST*, PT); (2) removing firms in financial and insurance sectors; (3) omitting observations with severely abnormal/missing data or

fewer than three years of records; (4) winsorizing continuous variables at 1 % and 99 % to mitigate outlier effects. The final dataset comprises 8055 firm-year observations. Firm characteristics and financial metrics are sourced from Wind and CSMAR databases.

4.2. Variable definition and measurement

4.2.1. Explained variable: organizational resilience

Measurement of organizational resilience can be approached through process, outcome, and capability perspectives (Xie and Chen, 2022), with the capability-based view best capturing its essence. Drawing on Ortiz-de-Mandojana and Bansal's research (Ortiz-de-mandojana and Bansal, 2016), this study conceptualizes organizational resilience as a two-dimensional construct: financial volatility and sustainable performance growth. Financial volatility is measured by the standard deviation of monthly stock returns, and sustainable performance growth is quantified using three-year total revenue growth.

Firms with low financial volatility demonstrate sharper environmental sensing and adaptive agility, enabling them to buffer external shocks effectively. Additionally, those achieving high sustainable performance growth exhibit robust crisis resilience and opportunity-capturing capacities. These two metrics are then synthesized via the entropy weight method to create a composite resilience index, holistically assessing a firm's capacity to anticipate, adapt, recover, and grow.

4.2.2. Explanatory variables: ESG greenwashing

Measuring ESG greenwashing (GW) poses challenges due to its inherent opacity and dynamic manifestations, leading researchers to rely on indirect proxies. Current approaches include consumer surveys (Chen and Chang, 2013), scenario analysis (Fella and Bausa, 2024), and experimental designs (Sajid et al., 2024) to detect greenwashing patterns. Scholars like Huang et al. (2020) have quantified GW intensity by differentiating selective environmental disclosures from symbolic ones through index construction. In addition, advances in machine learning enable more granular measurements, exemplified by Wang et al. (2024c) using Bayesian machine learning to study GW's financing impacts.

This study adopts Zhang's (Zhang, 2023b) methodology, calculating ESG greenwashing as the normalized discrepancy between Bloomberg ESG disclosure scores (reflecting stated commitments) and Huazheng ESG performance scores (assessing actual outcomes). The Bloomberg ESG score relies on the completeness and transparency of ESG information disclosed publicly by enterprises, while the Huazheng ESG rating is a rating system customized based on China's policies and industry characteristics. It integrates indicators such as environmental violations and safety incidents, and dynamically tracks negative ESG events of enterprises, which can better reflect the true performance of enterprises

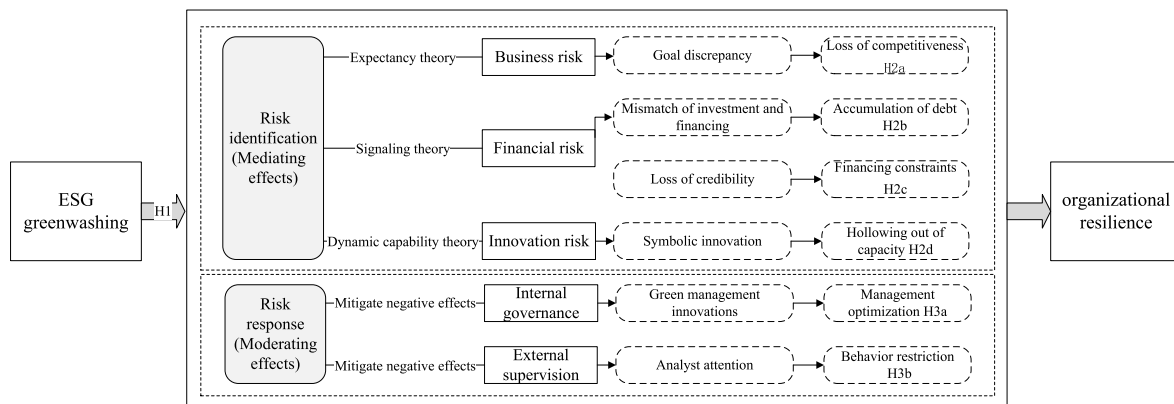


Fig. 1. A theoretical framework for ESG greenwashing acting on organizational resilience.

under the local regulatory framework. Thus, the discrepancy between the two can reflect the level of ESG greenwashing.

The ESG greenwashing index (GW) is scaled between 0 and 1 using min-max normalization, where higher values indicate stronger greenwashing tendencies. The measurement formula is specified in equation (1).

$$GW_{i,t} = \left(\frac{ESGdisclosure_{i,t} - \bar{E}SGdisclosure_{i,t}}{\sigma ESGdisclosure_{i,t}} \right) - \left(\frac{ESGperformance_{i,t} - \bar{E}SGperformance_{i,t}}{\sigma ESGperformance_{i,t}} \right) \quad (1)$$

$ESGdisclosure_{i,t}$ and $ESGperformance_{i,t}$ represents the corporate disclosure score and the actual performance score, respectively, $\bar{E}SGdisclosure_{i,t}$ and $\bar{E}SGperformance_{i,t}$ are the mean of the two. $\sigma ESGdisclosure_{i,t}$ and $\sigma ESGperformance_{i,t}$ are the standard deviations of the two. $GW_{i,t}$ is the measured level of ESG greenwashing.

4.2.3. Control variables

Drawing on the experience of existing studies, this study selects control variables from four aspects: enterprise characteristics, financial condition, operating condition, and corporate governance, to avoid the interference of other possible factors on the regression results. Among them, firm characteristics include firm age (Age) and firm size (Size). Financial condition includes asset-liability ratio (Lev), fixed asset ratio (Fixed), and book-to-market ratio (BM). Operating condition includes net profit margin of total assets (Roa), return on net assets (Roe), asset turnover ratio (Ato), and corporate growth (Growth). Corporate governance includes the proportion of independent directors (Indep), management expense ratio (Mfee), and the proportion of shares held by the

top ten shareholders (Top10). The specific measures and descriptions of the variables are shown in Table 1.

4.3. Empirical model design

4.3.1. Baseline regression model

To test the impact of ESG greenwashing on organizational resilience, this study constructs the following baseline model (Eq. (2)).

$$Resilience_{i,t} = \alpha + \beta GW_{i,t} + \delta Controls_{i,t} + \sum Year + \sum Firm + \varepsilon_{i,t} \quad (2)$$

Where $Resilience_{i,t}$ is the explanatory variable of organizational resilience; $GW_{i,t}$ is the explanatory variable of ESG greenwashing level; $Controls_{i,t}$ is the control variable; $Year$ and $Firm$ are the time effect and individual effect, respectively, and $\varepsilon_{(i,t)}$ is the random disturbance term.

4.3.2. Mediating effect model

To explore the mediating mechanisms of ESG greenwashing affecting organizational resilience, this study constructs a test model based on theories and concerning the suggestion of Jiang (2022) on the analysis of mechanism (Eq. (3)):

$$M_{i,t} = \alpha + \beta GW_{i,t} + \delta Controls_{i,t} + \sum Year + \sum Firm + \varepsilon_{i,t} \quad (3)$$

In this model, $M_{i,t}$ represents the three mediating mechanism variables of business risk, financial risk, and innovation risk. The rest of the

Table 1
Variable descriptions.

Types	Variable name	Symbol	Measurement methods	Source
Dependent variable	Organization resilience	Resilience	Based on the volatility of stock returns and the growth of cumulative sales revenue, the entropy weight method is used to calculate the comprehensive score	CSMAR
Independent variable	ESG greenwashing	GW	Difference between standardized ESG disclosure score and ESG performance score	Bloomberg and Wind
Mechanism variables	Lerner index	LI	(Operating income - operating cost - selling expense - administrative expense)/operating income	CSMAR
	Kaplan-Zingales index	KZ	Kaplan-Zingales index (see Appendix A)	CSMAR
	Default distance	DD	Default distance (see Appendix A) (Ding et al., 2021)	CSMAR
	New quality productive forces	NPQ	Construct an evaluation index system and calculate it using the entropy method (see Appendix A) (Li et al., 2024)	CSMAR
Moderating variables	Green management innovation	Gm	Sum up five indicators: ISO14001 certification (obtained or not), ISO9001 certification (obtained or not), the environmental management system, environmental education and training, and environmental special actions. (Xi and Zhao, 2022)	CSMAR
Control variables	Analyst attention	AnA	ln(The total number of analysts tracking the enterprise within one year + 1)	CSMAR
	Firm age	Age	ln(Year of the year - year of establishment + 1)	CSMAR
	Firm size	Size	Natural logarithms of total assets	CSMAR
	Leverage	Lev	Total liabilities at year - end divided by total assets at year - end	CSMAR
	Fixed asset ratio	Fixed	Ratio of fixed net assets to total assets	CSMAR
	Return of assets	Roa	Net profit/average balance of total assets	CSMAR
	Return of equity	Roe	Net profit/average balance of shareholders' equity	CSMAR
	Total assets turnover	Ato	Operating income/Average assets	CSMAR
	Revenue growth rate	Growth	(Current year's operating income/last year's operating income) - 1	CSMAR
	Book to market ratio	BM	Book value/total market value	CSMAR
	Proportion of independent directors	Indep	Independent directors divided by the number of directors	CSMAR
	Management expense ratio	Mfee	Administrative expenses divided by revenue	CSMAR
	Top 10 shareholders' shareholding ratio	Top10	The proportion of the shares held by the top ten shareholders of the firm	CSMAR

Note: CSMAR is the China Stock Market & Accounting Research Database, Bloomberg is Bloomberg L.P., and Wind is Wind Information Co. Ltd.

Table 2

Descriptive statistics of key variables.

Variable	Obs	Mean	p50	Std.Dev.	Min	Max
Resilience	8055	0.898	0.909	0.0574	0.0709	0.998
GW	8055	−0.475	−0.527	1.270	−5.546	5.691
Age	8055	2.980	3.045	0.319	1.099	3.761
Size	8055	23.42	23.34	1.284	19.57	26.45
Lev	8055	0.482	0.496	0.191	0.0349	0.927
Fixed	8055	0.240	0.199	0.180	0.0003	0.954
RoA	8055	0.0487	0.0406	0.0616	−0.382	0.255
Roe	8055	0.0859	0.0868	0.121	−0.962	0.415
Ato	8055	0.683	0.577	0.510	0.0182	7.788
Growth	8055	0.195	0.101	1.495	−0.875	84.99
BM	8055	1.598	0.973	1.973	0.0301	32.60
Indep	8055	0.377	0.364	0.0594	0.182	0.800
Mfee	8055	0.0705	0.0566	0.0625	0.0007	1.707
Top10	8055	0.604	0.609	0.161	0.127	1.012

parameters, control variables, and fixed effects are set to be consistent with the baseline regression model (Eq. (2)).

4.3.3. Moderating effect model

To verify the moderating effects of green management innovation and analyst attention, this paper constructs the following moderating effect models (Eq. (4) and Eq. (5)):

$$Resilience_{i,t} = \alpha + \beta_1 GW_{i,t} + \beta_2 GW_{i,t} \times Gm_{i,t} + \beta_3 Gm_{i,t} + \delta Contorls_{i,t} + \sum Year + \sum Firm + \varepsilon_{i,t} \quad (4)$$

$$Resilience_{i,t} = \alpha + \beta_4 GW_{i,t} + \beta_5 GW_{i,t} \times Ana_{i,t} + \beta_6 Ana_{i,t} + \delta Contorls_{i,t} + \sum Year + \sum Firm + \varepsilon_{i,t} \quad (5)$$

Where $GW_{i,t}$ represents green management innovation and $Ana_{i,t}$ represents the level of analyst attention. $GW_{i,t} \times Gm_{i,t}$ and $GW_{i,t} \times Ana_{i,t}$ are the interaction terms between ESG greenwashing and green management innovation and analyst attention respectively. β_2 and β_5 represent the coefficients of the mediating effects of green management innovation and analyst attention.

5. Empirical results

5.1. Descriptive statistics

Table 2 reports descriptive statistics for key variables. The organizational resilience indicator exhibits a right-skewed distribution, with a standard deviation of 0.057, suggesting limited overall volatility. However, its extreme values reveal substantial structural disparities in crisis response capabilities across firms. For ESG greenwashing (GW), the distribution shows a polarization pattern: while the sample mean (−0.475) and median (−0.527) indicate generally low GW levels, the large standard deviation (1.270) and extreme values confirm significant heterogeneity. Notably, the resilience metric's right skew aligns with the sample's high baseline risk resistance, whereas GW's high dispersion implies potential opportunistic environmental disclosure strategies among some firms. These patterns provide critical empirical grounding for subsequent heterogeneity analyses.

Additionally, the correlation coefficient (see Appendix B) reveals that ESG greenwashing has a coefficient of −0.040 with organizational resilience, providing preliminary support for Hypothesis 1.

5.2. Baseline regression results

Table 3 presents the direct effects of ESG greenwashing (GW) on organizational resilience. Column (1) controls for firm and year fixed effects alone, while Columns (2) through (5) progressively incorporate

control variables spanning enterprise characteristics, financial condition, operating condition, and corporate governance dimensions. The results demonstrate that GW exhibits statistically significant negative impacts on resilience at the 1 % level, regardless of control variable inclusion. This consistency robustly supports Hypothesis 1 that ESG greenwashing undermines organizational resilience.

5.3. Endogenous treatment

5.3.1. Propensity score matching method

This study utilizes propensity score matching to reduce the impact of possible sample self-selection issues on the regression outcomes. It divides the sample into a treatment group (i.e., firms with a high greenwashing level) and a control group (i.e., firms with a low greenwashing level) by the yearly average of ESG greenwashing in the industry. Concurrently, this study chooses control variables as corresponding variables and respectively uses nearest-neighbor matching, radius matching and kernel matching to reevaluate the regression model, with the outcomes displayed in columns (1), (2), and (3) of Table 4. The results show that the coefficients are still significantly negative.

5.3.2. Two-stage least squares regression (2SLS)

Potential reverse causality between ESG greenwashing and organizational resilience may exist in uncertain environments, as firms with weaker resilience could be incentivized to mask their vulnerabilities through ESG greenwashing. To mitigate the problem of endogeneity, this study evaluates the average yearly greenwashing in the same industry and province (Zhang et al., 2022b) as the instrumental variables, with results presented in Table 4. The first-stage regression results in columns (4) and (6) indicate that the instrumental variables meet the correlation requirements. More importantly, as reported in columns (5) and (7), the second-stage regression results show that the coefficients of ESG greenwashing are significantly negative at the 5 % level. In addition, the values of the Kleibergen-Paap rk LM statistics are significant at the 1 % level, and the values of the Kleibergen-Paap Wald F statistics reject the hypothesis of weak instrumental variables, indicating that instrumental variables are feasible. Overall, the main results remain robust.

5.3.3. System Generalized Method of Moments (SYS-GMM)

Compared to the two-way fixed effects model, the SYS-GMM model more effectively addresses endogeneity issues arising from omitted variables or reverse causality. This study employs the SYS-GMM model to test potential endogeneity, with results presented in Column (8) of Table 4. The estimated coefficients for both the first-order lag terms of organizational resilience (L.Resilience) and ESG greenwashing (GW) are statistically significant at the 1 % level. Furthermore, the SYS-GMM model passes the AR(2) and Hansen tests, indicating that the null hypothesis of second-order autocorrelation in the disturbance term is rejected at the 10 % significance level, and the instrumental variables satisfy exogeneity. Thus, these results confirm the robustness of the baseline regression findings.

5.4. Robustness test

This study also conducts the following robustness tests to guarantee the validity of the main findings.

5.4.1. Replacement of the ESG greenwashing indicator

Following the measurements of Hu (Hu et al., 2023) and Li (Li et al., 2025c), this study reconstructs the ESG greenwashing (GW) metric using equation (6). Specifically, it normalizes both disclosure scores and actual performance by subtracting their minimum values and scaling them by the range between maximum and minimum values. Column (1) of Table 5 shows that the coefficient remains statistically significant at the 1 % level under this alternative measurement, providing further

Table 3

Estimation results of ESG greenwashing impact on firm resilience.

Variables	(1)	(2)	(3)	(4)	(5)
	Resilience	Resilience	Resilience	Resilience	Resilience
GW	−0.0011*** (0.0003)	−0.0011*** (0.0003)	−0.0011*** (0.0003)	−0.0011*** (0.0003)	−0.0011*** (0.0003)
Age		0.0188** (0.0080)	0.0237*** (0.0080)	0.0238*** (0.0080)	0.0282*** (0.0082)
Size		0.0018* (0.0010)	−0.0004 (0.0011)	0.0002 (0.0012)	−0.0004 (0.0012)
Lev			−0.0258*** (0.0040)	−0.0306*** (0.0045)	−0.0301*** (0.0045)
Fixed			−0.0016 (0.0046)	−0.0036 (0.0047)	−0.0032 (0.0048)
BM			0.0069*** (0.0005)	0.0066*** (0.0005)	0.0067*** (0.0005)
Roa				−0.0471* (0.0254)	−0.0474* (0.0253)
Roe				0.0115 (0.0117)	0.0105 (0.0117)
Ato				0.0010 (0.0019)	0.0006 (0.0020)
Growth				−0.0024* (0.0012)	−0.0030** (0.0012)
Indep					−0.0004 (0.0083)
Mfee					−0.0195 (0.0144)
Top10					0.0177*** (0.0048)
Constant	0.8984*** (0.0001)	0.8008*** (0.0328)	0.8385*** (0.0335)	0.8282*** (0.0351)	0.8214*** (0.0373)
Firm FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
R ²	0.835	0.836	0.845	0.845	0.846
N	8055	8055	8055	8055	8055

Note: *p < 0.10, **p < 0.05, ***p < 0.01; Cluster-robust standard errors in parentheses.

Table 4

Endogenous issue treatment results of ESG greenwashing impact on organizational resilience.

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	PSM			2SLS				SYS - GMM
	Resilience	Resilience	Resilience	GW	Resilience	GW	Resilience	Resilience
L.Resilience								0.1450*** (0.0303)
GW	−0.0011*** (0.0003)	−0.0011*** (0.0003)	−0.0011*** (0.0003)		−0.0073*** (0.0017)		−0.0043** (0.0017)	−0.0013*** (0.0005)
IV1_GW				0.8725*** (0.0671)				
IV2_GW						0.8782*** (0.0689)		
Controls	Control	Control	Control	Control	Control	Control	Control	Control
Constant	0.8332*** (0.0350)	0.8331*** (0.0350)	0.8331*** (0.0350)					
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
K-P Wald F				168.87		126.64		
K-P rk LM					91.63***		114.47***	
AR(1)								0.000
AR(2)								0.964
Hansen test								0.161
R ²	0.847	0.847	0.847		0.843		0.851	
N	8035	8049	8049	8055	8055	8055	8055	5509

Note: *p < 0.10, **p < 0.05, ***p < 0.01; Cluster-robust standard errors in parentheses.

validation for Hypothesis 1.

$$GW_{it} = \left(\frac{ESGdisclosure_{it} - \min ESGdisclosure_{it}}{\max ESGdisclosure_{it} - \min ESGdisclosure_{it}} \right) - \left(\frac{ESGperformance_{it} - \min ESGperformance_{it}}{\max ESGperformance_{it} - \min ESGperformance_{it}} \right) \quad (6)$$

5.4.2. Modifying the cluster-robust standard errors

Clustering robust standard errors at higher levels can effectively reduce the effect of heterogeneity among observed individuals at lower levels and mitigate the bias of the regression model, so this study clusters robust standard errors at the industry-year two-dimensional level. Column (2) of Table 5 illustrates the robustness of the results.

5.4.3. Subsample regression

Considering that the shock of the unexpected event of the COVID-19 pandemic in 2020 may have an impact on the organizational resilience of firms, the regression is conducted by excluding the sample data after 2020, and the coefficients in Column (3) of Table 5 are significant at the 10 % level, which again verifies the robustness of the regression results.

5.4.4. Multidimensional fixed effects model (FEM)

The baseline regression initially controlled only for firm and time-fixed effects. However, to address potential confounding from unobserved industry and city-level factors, this study enhances the model by including industry and city-fixed effects. As shown in Column (4) of Table 5, the coefficient on ESG greenwashing remains statistically significant, confirming the robustness of our findings regarding its adverse impact on organizational resilience.

6. Mechanism analysis

6.1. Analysis of the mediating effects

The results of baseline regression after a series of endogeneity and robustness tests show that ESG greenwashing significantly reduces the level of corporate organizational resilience. This section utilizes mechanism testing to analyze how ESG greenwashing weakens organizational resilience by influencing different types of idiosyncratic risks.

6.1.1. Business risk mechanism

Firms with dominant product market positions exhibit greater

resilience against external shocks in dynamic environments. Their market power enables them to partially offset adverse impacts by transferring shocks to other stakeholders, thereby stabilizing cash flows. Additionally, competitive leadership can reduce information asymmetry, facilitating more accurate market assessments and risk mitigation. However, ESG greenwashing exposes firms to business risks. First, market share erosion undermines resource integration capabilities, constraining R&D investments and technological upgrading. Second, diminished competitiveness reduces organizational agility, impairing firms' ability to adapt to market changes and capture emerging opportunities. Following Jia and Wei (2019), this study measures competitive position using the Lerner index (LI), where higher values indicate stronger pricing power and risk resilience. As reported in Column (1) of Table 6, the significant coefficient of -0.0016 ($p < 0.05$) confirms that ESG greenwashing erodes market competitiveness, exacerbates business risks, and ultimately weakens organizational resilience, thereby validating Hypothesis 2a.

6.1.2. Financial risk mechanism

Financing capacity serves as a critical determinant for corporate survival and sustainable growth, directly influencing organizational resilience. Firms with diversified financing channels typically enjoy lower financing thresholds, reduced capital costs, and sufficient liquidity to withstand adverse shocks. However, the short-sighted nature of ESG greenwashing may escalate debt overhang and default risks, exposing firms to direct financial penalties, indirect reputational costs, and legal sanctions. To validate this financial risk mechanism, this study considers both debt default and financing constraints, drawing on the research of Ding et al. (2021): debt default probability (DD), where lower values indicate higher default risk, and the KZ index measuring financing constraints, with higher values reflecting tighter capital access. Column (2) of Table 6 shows a significant coefficient ($\beta = -0.1051$, $p < 0.01$), confirming that ESG greenwashing reduces the default distance. Simultaneously, the positive coefficient ($\beta = 0.0311$, $p < 0.05$) in Column (3) demonstrates that greenwashing intensifies financing constraints. These findings collectively verify that ESG greenwashing undermines organizational resilience by amplifying debt default risks and financing constraints, supporting Hypotheses 2b and 2c.

6.1.3. Innovative risk mechanism

New quality productive forces, as the core driver enhancing corporate competitiveness and sustainable development, integrate green innovation, technological advancement, and digital transformation (Li et al., 2024) (Wei et al., 2025). They embody sustainable growth principles while aligning with high-quality development imperatives, serving as the cornerstone for renewing dynamic capabilities and energizing enterprises during digital transitions. However, ESG greenwashing may derail this process through opportunistic motives, diverting resources from substantive technological upgrades to superficial image-building. Such practices conflict with the essence of new quality productive forces and incur long-term developmental costs. Following Li's method (Li et al., 2024), this research constructs a new quality productive forces (NPQ) index to assess innovation risks. The significantly negative coefficient in Table 6, Column (4) ($\beta = -0.6408$, $p < 0.01$) reveals that greenwashing suppresses productive forces. Formalistic environmental investments crowd out technological funds, while disingenuous disclosures distort organizational learning, trapping green innovation and digital transformation into low-level equilibria. This ultimately erodes dynamic capabilities, validating Hypothesis 2d.

6.2. Analysis of the moderating effects

To verify the moderating effects of green management innovation and analyst attention on the negative impacts of ESG greenwashing, this study conducts a moderating effect analysis and constructs marginal effect graphs of ESG greenwashing on organizational resilience (Fig. 2).

Table 5
Robustness test results of ESG greenwashing impact on organizational resilience.

Variables	(1) Change GW measurement	(2) Modifying cluster-robust standard errors	(3) Subsample regression	(4) Multidimensional FEM
	Resilience	Resilience	Resilience	Resilience
GW2	-0.0025** (0.0010)			
GW		-0.0012*** (0.0003)	-0.0006* (0.0003)	-0.0011*** (0.0003)
Controls	Control	Control	Control	Control
Constant	0.8196*** (0.0372)	0.8206*** (0.0365)	0.8016*** (0.0441)	0.8214*** (0.0367)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Industry FE	No	No	No	Yes
City FE	No	No	No	Yes
R ²	0.845	0.843	0.909	0.846
N	8055	8055	5593	8055

Note: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$; Cluster-robust standard errors in parentheses.

Table 6
Mechanism analysis of ESG greenwashing impact on organizational resilience.

Variables	(1)	(2)	(3)	(4)	(5)	(6)
	Mediating effects				Moderating effects	
	LI	DD	KZ	NPQ	Resilience	Resilience
GW	−0.0016** (0.0007)	−0.1051*** (0.0142)	0.0311** (0.0398)	−0.6408*** (0.0619)	−0.0020*** (0.0004)	−0.0012*** (0.0003)
Gm × GW					0.0021** (0.0008)	
AnA × GW						0.0006** (0.0003)
Gm					0.0015 (0.0012)	
AnA						−0.0002 (0.0005)
Controls	Control	Control	Control	Control	Control	Control
Constant	−0.0206 (0.1182)	−1.4845 (1.7528)	1.8399 (4.1206)	−13.0825 (8.8497)	0.8251*** (0.0371)	0.8181*** (0.0379)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
R ²	0.875	0.678	0.782	0.697	0.846	0.846
N	8055	8055	8055	8033	8055	8055

Note: *p < 0.10, **p < 0.05, ***p < 0.01; Cluster-robust standard errors in parentheses.

This investigation provides actionable insights for mitigating greenwashing risks while offering practical strategies for effective governance, thereby enabling firms to address ESG credibility challenges through enhanced managerial practices and market oversight.

6.2.1. Green management innovations

This study evaluates green management innovation (Gm) using Xi's (Xi and Zhao, 2022) method and constructs its interaction term with ESG greenwashing to verify its moderating effect. As shown in Column (5) of Table 6, the interaction term (Gm × GW) has a significant coefficient ($\beta = 0.0021$, $p < 0.05$). Furthermore, in Fig. 2, with a high level of green management innovation, firms' ESG greenwashing will have a smaller marginal negative impact on organizational resilience, thereby validating Hypothesis 3a. This is because systematic green management practices enhance organizational sensitivity to external dynamics and internal risks through technological and procedural innovations, enabling firms to anticipate the consequences of greenwashing and proactively implement governance measures. Additionally, environmental training, clean technology R&D, and supply chain optimization improve resource reallocation efficiency (Wu and Pan, 2022), strengthening adaptive capacity during periods of greenwashing exposure. These mechanisms collectively facilitate alignment with green transition goals and bolster sustainable development strategies.

6.2.2. Analyst attention

Following Kong (Kong et al., 2013), this study measures analyst coverage (AnA) as the natural logarithm of the total number of analysts tracking a firm plus one and constructs its interaction term with ESG greenwashing. As demonstrated in Column (6) of Table 6, the interaction term (AnA × GW) shows statistical significance ($\beta = 0.0006$, $p < 0.05$). Fig. 2 shows that increased analyst attention will reduce the negative marginal impact of ESG greenwashing, confirming Hypothesis 3b. This moderating effect operates primarily through two mechanisms: first, continuous analyst monitoring creates market surveillance pressure that constrains firms' opportunistic environmental disclosure strategies for impression management; second, analysts reduce information asymmetry between investors and management by decoding green innovation pathways and ESG risks, making greenwashing practices more detectable and triggering market disciplinary mechanisms. Such dual governance effects not only enhance the authenticity of environmental disclosures but also compel firms to shift from passive compliance to proactive adaptation through market constraints and reputational incentives, ultimately strengthening organizational resilience via improved environmental governance efficacy.

7. Heterogeneity analysis

Empirical evidence confirms the significant negative impact of ESG greenwashing on organizational resilience. However, this effect

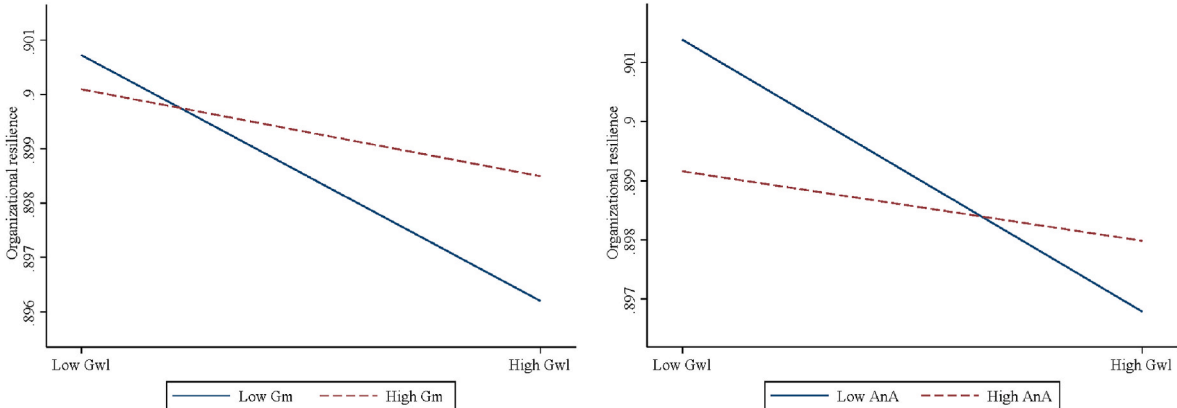


Fig. 2. Moderating interaction diagrams of green management innovations and analyst attention.

demonstrates heterogeneity contingent on firm-specific characteristics and contextual environments. To systematically examine these divergent impacts, this study conducts grouped regression analyses across three dimensions: corporate characteristics, internal governance quality, and external policy environment. This multi-perspective approach enables comprehensive identification of how ESG greenwashing differentially influences organizational resilience under varying environments.

7.1. Corporate characteristics

7.1.1. Ownership

This study categorizes firms into state-owned enterprises (SOEs) and non-SOEs for grouped regression. Table 7, Columns (1) and (2) indicate that ESG greenwashing significantly weakens organizational resilience in SOEs ($\beta = -0.0015$, $p < 0.01$), while its effect remains statistically insignificant for non-SOEs. This disparity stems from dynamic differences in institutional constraints and response mechanisms. As key implementers of national economic strategies, SOEs operate under dual institutional identities as policy instruments and market entities. Their environmental misconduct triggers stronger public scrutiny due to their privileged policy resources, exacerbating public disillusionment and eroding organizational credibility. In contrast, non-SOEs leverage market-oriented governance structures and flexible crisis management, employing adaptive strategies like supply chain adjustments and technological optimization to mitigate ESG-related reputational shocks. This organizational agility enables non-SOEs to maintain resilience thresholds despite greenwashing incidents.

7.1.2. Industry type

This study classifies enterprises into high-tech and non-high-tech industries based on two authoritative Chinese regulatory frameworks: the *Guidelines for Industry Classification of Listed Companies* (2012 Edition) and the *National Key Supported High-Tech Fields*, ensuring alignment with China's official industry standards and strategic technology development priorities. As shown in Table 7, Columns (3) and (4), ESG greenwashing significantly reduces organizational resilience in non-high-tech firms ($\beta = -0.0015$, $p < 0.01$), while the coefficient for high-tech firms (-0.0006) remains statistically insignificant. This divergence stems from high-tech firms' greater sensitivity to technological changes and stronger innovation capabilities. In the digital era, these firms leverage big data, cloud computing, and AI to drive dual transformations (digital and green), establishing smart monitoring systems and dynamic capabilities. Such technological infrastructure enables real-time environmental risk detection and strategic adjustments, enhancing crisis responsiveness. Additionally, digitally enabled collaborative networks expand resource accessibility, creating resilient ecosystems that mitigate resource shortages caused by environmental

crises, thereby reducing the negative effects of ESG greenwashing.

7.1.3. Corporate life cycle

Applying the corporate life cycle theory, which conceptualizes firms as evolving organisms with distinct governance environments, performance patterns, and risk exposures across the growth stage. This study employs Li's classification framework (Li et al., 2011) to categorize firms into growth, maturity, and decline phases, utilizing a composite scoring system that integrates four key indicators: sales revenue growth rate, retained earnings, capital expenditures, and firm age. Columns (5), (6) and (7) in Table 7 demonstrate that ESG greenwashing significantly undermines organizational resilience in mature-phase firms ($\beta = -0.0015$, $p < 0.01$) and decline-phase firms ($\beta = -0.0013$, $p < 0.01$), with stronger effects observed in mature-stage enterprises. This pattern may arise because growth-phase firms, constrained by rigid financing needs and underdeveloped brand equity, exhibit limited systemic incentives for ESG greenwashing and attract less stakeholder scrutiny. In contrast, mature-phase firms' standardized governance structures and stable market positions make their brand reputations vulnerable to ESG rating declines caused by greenwashing, triggering trust breakdowns. Meanwhile, decline-phase firms suffer from diminished slack resources and delayed strategic adjustments, jointly weakening their crisis management capacity when ESG greenwashing exposures occur.

7.2. Internal governance environments

To examine heterogeneity in internal governance environments, the study follows Yan's (Yan et al., 2023) method by stratifying the sample into two groups based on whether there are internal control weaknesses. As shown in columns (1) and (2) of Table 8, firms with stronger internal governance environments (no control weaknesses) exhibit a weaker negative impact of ESG greenwashing on organizational resilience. This advantage stems from robust management oversight and checks-and-balances mechanisms, which enhance decision-making rationality to facilitate timely risk identification and mitigation and strengthen resilience against major disruptions. Simultaneously, these mechanisms enforce regulatory compliance, curbing ESG greenwashing practices and preserving corporate legitimacy. Therefore, establishing sound internal control systems serves as a critical institutional safeguard for firms to manage post-exposure ESG greenwashing crises and enhance organizational resilience in complex environments.

7.3. External policy environments

The heterogeneity in regional environmental regulatory intensity moderates the impact of ESG greenwashing on organizational resilience through differential compliance pressures, production costs, and pollution-reduction incentives (Song and Gao, 2025). Referring to the

Table 7
Heterogeneity test for corporate characteristics.

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	SOEs	Non-SOEs	High-tech	Non-high-tech	Growth	Maturity	Decline
	Resilience	Resilience	Resilience	Resilience	Resilience	Resilience	Resilience
GW	-0.0015*** (0.0004)	-0.0006 (0.0004)	-0.0006 (0.0005)	-0.0015*** (0.0004)	-0.0005 (0.0010)	-0.0015 *** (0.0005)	-0.0013 *** (0.0005)
Controls	Control	Control	Control	Control	Control	Control	Control
Constant	0.7902*** (0.0542)	0.8355*** (0.0537)	0.7951*** (0.0562)	0.8151*** (0.0515)	0.7057 *** (0.1366)	0.7559 *** (0.0590)	1.0117 *** (0.0735)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R2	0.854	0.836	0.848	0.847	0.891	0.872	0.834
N	4361	3672	3189	4874	915	2563	3825
Between-group difference test	0.048**		0.084*				

Note: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$; Cluster-robust standard errors in parentheses. Between-group difference tests are computed using Fisher's permutation method to analyze whether there are significant distribution differences among different groups in the binary classification results.

Table 8
Heterogeneity test for internal and external environments.

Variables	(1)	(2)	(3)	(4)
	Defective	Non-defective	Low regulation	High regulation
	Resilience	Resilience	Resilience	Resilience
GW	−0.0011** (0.0004)	−0.0018*** (0.0004)	−0.0010** (0.0004)	−0.0018*** (0.0004)
Controls	Control	Control	Control	Control
Constant	0.8490*** (0.0483)	0.8502*** (0.0668)	0.8584*** (0.0566)	0.8288*** (0.0562)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
R ²	0.842	0.842	0.683	0.843
N	4641	3173	3773	3863
Between-group difference test	0.096*		0.080*	

Note: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$; Cluster-robust standard errors in parentheses. Between-group difference tests are computed using Fisher's permutation method to analyze whether there are significant distribution differences among different groups in the binary classification results.

method of Shao (Shao et al., 2024), this study measures regulatory intensity using the ratio of environmental-related keywords to total words in regional government reports. Firms are categorized into low- and high-regulation groups based on annual median intensity. Column (4) of Table 8 shows that in high-regulation regions, the coefficient of GW is −0.0018 ($p < 0.01$), significantly stronger than the −0.0010 ($p < 0.05$) in low-regulation regions, indicating that stricter environmental regulations amplify the erosion of resilience by ESG greenwashing. Because dynamic environmental standards and disclosure requirements impose binding constraints, greenwashing practices like false reporting or data manipulation incur direct penalties and operational disruptions from litigation or remediation, weakening enterprises' risk prevention capability. Additionally, short-term compliance strategies cause strategic resource misallocation, hindering firms' ability to balance immediate regulatory costs with long-term risks, thereby accelerating resilience depletion. While strict regulations curb overt violations in the short run, they may incentivize firms to divert environmental investments into greenwashing tactics rather than innovation. This strategic response prevents cost-effective emission reductions through technological upgrades and erodes adaptive capacity to external shocks. Policymakers should therefore integrate regulatory enforcement with innovative incentives, such as tax preference and green finance, to redirect corporate environmental spending from passive compliance to proactive capability-building, achieving synergies between environmental performance and organizational resilience.

7.4. Level of organizational resilience

To more comprehensively assess how ESG greenwashing affects organizational resilience across varying resilience levels while mitigating outlier distortions, this study employs a panel quantile regression model at six quantiles (15%, 30%, 40%, 60%, 70%, 85%). The empirical results combined with Table 9 and Fig. 3 show statistically significant negative coefficients ($P < 0.01$) for ESG greenwashing across all quantiles, robustly supporting Hypothesis 1. Notably, the marginal impact of ESG greenwashing on organizational resilience reaches its peak at around 30% percentile and then shows a monotonically decreasing trend with the increase of the percentile. This pattern stems from differences in resource endowments and risk mitigation capacities. For low-resilience firms (< 30%), short-term gains from ESG greenwashing offset compliance costs, creating reverse incentives where inferior practices prevail. In mid-resilience firms (30%–60%), improved governance mechanisms expose hidden costs of greenwashing, amplifying its harm. High-resilience firms (> 70%) leverage redundant resource

reserves and intelligent risk-control systems to reduce market penalties, diluting greenwashing's consequences. These findings suggest tailored anti-greenwashing strategies: low-to-mid resilience firms should prioritize authentic ESG disclosure audits, while high-resilience firms must guard against complacency-driven regulatory arbitrage tendencies.

8. Conclusions, implications and limitations

8.1. Conclusions

Corporate organizational resilience, as a core capability for navigating environmental uncertainties, forms the micro-foundation of economic resilience systems. This study uses an integrated theoretical framework to examine the impact mechanisms of ESG greenwashing using data from China's A-share listed enterprises (2012–2022). The findings show that ESG greenwashing undermines organizational resilience through symbolic compliance and substantive avoidance behaviors, redirecting too many resources to superficial compliance efforts. Ultimately, this reduces competitiveness, tightens financial limitations, and hinders innovative capacity by undermining risk management systems and driving away investments in technological innovation. Notably, green management innovation has an institutional restructuring effect by establishing an environmental risk whole-process control system to optimize resource allocation, whereas analyst oversight strengthens constraints on greenwashing through improved information interpretation and market discipline. Heterogeneity analysis reveals that different enterprise attributes will form differentiated ESG greenwashing resistance capabilities. For instance, the agile decision-making structure and market-oriented operation mechanism of non-state-owned enterprises, the innovation-oriented organizational culture of high-tech enterprises, the strategic flexibility advantage of growth-stage enterprises, and the risk early warning system of enterprises with strong internal control. All can help enterprises better reduce the impact of ESG greenwashing risks on organizational resilience. However, intensified environmental regulations may trigger strategic compliance responses, exacerbating ESG greenwashing tendencies under combined compliance pressure and technological gaps, ultimately obstructing the development of dynamic capabilities. Further investigation reveals that the detrimental impacts of ESG greenwashing initially escalate but later diminish as organizational resilience shifts from low to high levels. This indicates that high-resilience organizations manage risks by employing redundant resources and activating learning mechanisms. These findings provide theoretical criteria for identifying ESG greenwashing risks and policy implications for building resilience enhancement mechanisms that balance compliance enforcement with capability cultivation, urging strengthened environmental disclosure audits for low-resilience firms and vigilance against compliance-driven regulatory arbitrage among high-resilience entities.

8.2. Practical implications

To harness ESG greenwashing and improve organizational resilience, this study delineates some pragmatic policy suggestions and development visions.

First, it's vital to improve environmental disclosure systems. On the one hand, adopt international standards (e.g., Global Reporting Initiative, Sustainability Accounting Standards Board) to standardize accounting methods, ensuring data comparability through unified metrics. Mandate disclosure of key indicators (e.g., carbon intensity, pollution levels) and compliance incidents, with industry-specific requirements. On the other hand, establish a multi-agency oversight framework led by environmental authorities, utilizing a shared data platform to integrate ecological, market, and administrative data (e.g., tax, energy). Implement annual randomized audits with cross-departmental verification and deploy anomaly detection algorithms to identify discrepancies between environmental investments and financial performance, enforcing

Table 9
Heterogeneity test for the level of organizational resilience.

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	OLS	Quantile regression					
	Resilience	15%	30%	40%	60%	70%	85%
GW	−0.0011*** (0.0002)	−0.0008*** (0.0000)	−0.0023*** (0.0000)	−0.0011*** (0.0001)	−0.0011*** (0.0001)	−0.0010*** (0.0000)	−0.0006*** (0.0000)
Controls	Control	Control	Control	Control	Control	Control	Control
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	8055	8055	8055	8055	8055	8055	8055

Note: *p < 0.10, **p < 0.05, ***p < 0.01; Cluster-robust standard errors in parentheses.

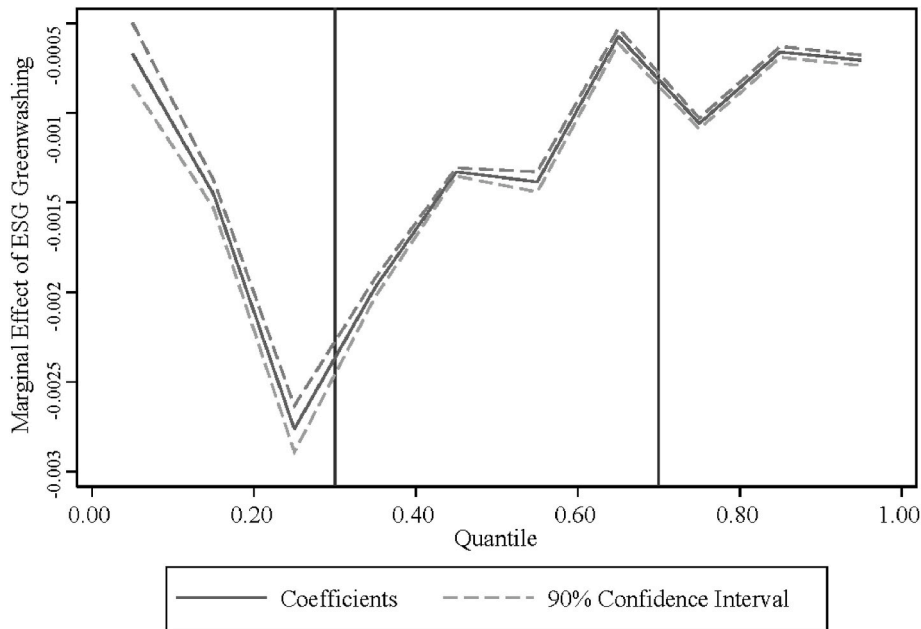


Fig. 3. Marginal effect of ESG greenwashing under different organizational resilience quantiles.

tiered penalties for falsification. This approach ensures both standardization and regulatory coordination.

Second, to enhance ESG governance, enterprises must prioritize digital transformation by incorporating blockchain traceability and AI monitoring into their environmental disclosure systems. This integration allows for dynamic risk-warning networks that employ intelligent algorithms to detect ESG greenwashing patterns. As a result, companies can shift from merely ensuring compliance to actively anticipating risks, helping them identify inconsistencies between their operational practices and the sustainability metrics reported. By automatically detecting anomalies in emissions data and supply chain information, organizations can proactively mitigate potential greenwashing risks before they lead to regulatory penalties or harm.

Third, implement a green management innovation incentive mechanism to guide businesses in creating customized transformation strategies that consider their industry traits and resource availability. This approach will focus on the complementary development of enhanced operational efficiency and reduced ecological footprints through adopting cleaner production techniques, exploring circular economy frameworks, and establishing a comprehensive environmental cost accounting system across the lifecycle. Companies need to integrate sustainability into their corporate culture by converting strategic environmental objectives into quantifiable performance indicators, thus creating self-regulating protections against ESG greenwashing.

Fourth, the capital market oversight framework needs restructuring by creating an analyst-engaged governance model. Businesses must

improve mandatory ESG disclosure practices by utilizing XBRL (eXtensible Business Reporting Language) taxonomies to improve the comparability of environmental data, alongside regular green performance updates to enable two-way dialogue with analysts. Regulatory bodies ought to implement competency evaluation systems for analysts, focusing on their accuracy in identifying ESG risks and detecting greenwashing. Moreover, utilizing regulatory technology to monitor the quality of research reports in time is crucial, with potential suspensions for third-party institutions found to be colluding, thereby increasing the credibility of market disciplinary systems.

Fifth, a tiered regulatory framework for corporate ESG greenwashing risks is vital. For state-owned enterprises, Party organizations should take a more active role in ESG governance. Growth-stage companies should benefit from support systems such as green technology incubators and compliance assistance platforms. Mature companies face flexible regulatory mechanisms tying environmental performance to carbon allowance allocations. Those with weak internal controls must undergo third-party governance audits and adopt suspension protocols for unresolved ESG deficiencies. For technology-intensive industries, differentiated green innovation subsidy policies should be formulated, focusing on cultivating clean technology patent pools and circular economy business models. At the same time, the innovation mechanism of green financial products should be improved, and tools such as environmental performance-linked bonds should be used to guide capital to flow into substantive transformation projects.

Finally, local governments should create flexible regulatory

frameworks with built-in adaptability for ESG greenwashing governance, leveraging digital innovation to establish dynamic coordination mechanisms. This involves setting tiered environmental standards aligned with regional ecological carrying capacities and deploying smart monitoring devices for tracking corporate compliance data to prevent strategic greenwashing tactics arising from rigid, one-size-fits-all enforcement. Concurrently, advancing blockchain applications for carbon emission traceability and AI-powered environmental risk early-warning systems enhance pollution source monitoring and disclosure accuracy.

8.3. Limitations

Although this research reveals the mechanisms of ESG greenwashing on organizational resilience, this paper still has some limitations. First, for endogenous treatment, we use the average yearly ESG greenwashing in the same industry and province as the instrumental variables and obtain the expected results. However, because of peer effects, these instrument variables may still relate to enterprises' ESG greenwashing, lacking Exogeneity and exclusivity. Therefore, they may offer only a partial solution to endogeneity. Future studies could consider more robust instruments, such as industry regulatory pressure or public environmental concern.

CRediT authorship contribution statement

Zhaoxia Wu: Writing – review & editing, Supervision, Project administration, Funding acquisition. **Siyu Long:** Writing – review & editing, Writing – original draft, Software, Resources, Methodology, Formal analysis, Data curation, Conceptualization. **Yumeng Wang:** Writing – review & editing, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jenvman.2025.126395>.

Data availability

Data will be made available on request.

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